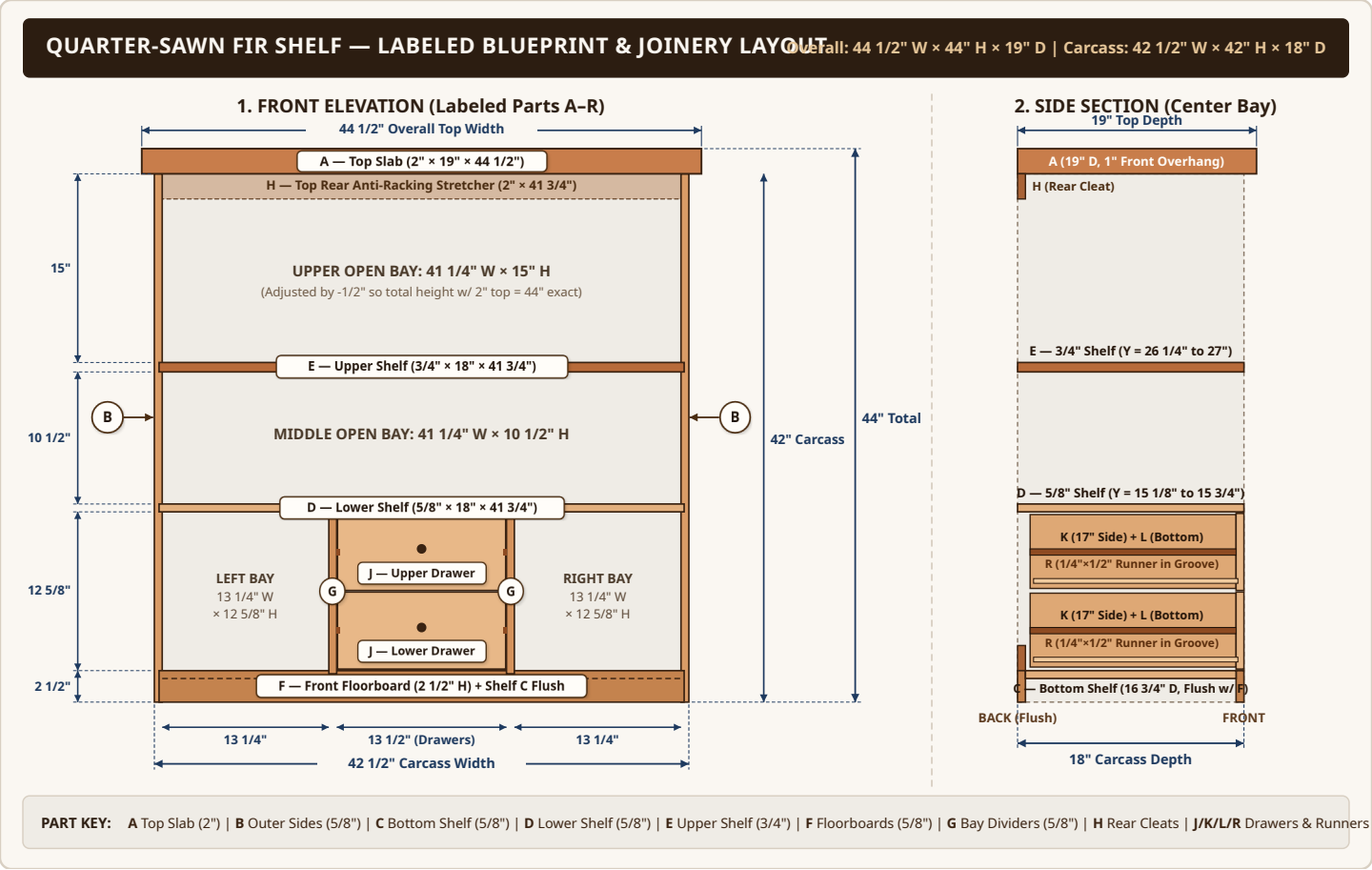
Quarter-Sawn Fir Shelf — Master Cut List & Labeled Shop Plan



1. Finished & Carcass Dimensions Summary

Dimension	Carcass (Box Only)	Overall (With 2" Top Installed)	Notes
Width	42 1/2" (42.500")	**44 1/2"** (44.500")	1" overhang on left and right sides
Height	42" (42.000")	**44"** (44.000")	Adjusted top opening to 15" so total height is 44" exact
Depth	18" (18.000")	**19"** (19.000")	1" overhang at front, flush at back

Vertical Height Stack-Up (Bottom to Top)

1. **Front & Rear Floorboards (F) + Flush Bottom Shelf (C):** 0" to **2 1/2"** (2.500")
2. **Lower 3-Quadrant Bay Opening (with 2 Center Drawers):** 2 1/2" to 15 1/8" (**12 5/8"** high)
3. **Lower Full-Width Shelf (D, 5/8" thick):** 15 1/8" to **15 3/4"** (0.625" thick)
4. **Middle Full-Width Open Bay:** 15 3/4" to 26 1/4" (**10 1/2"** high)
5. **Upper Full-Width Shelf (E, 3/4" thick):** 26 1/4" to **27"** (0.750" thick for zero sag)
6. **Upper Full-Width Open Bay:** 27" to **42"** (**15"** high)
7. **Overhanging Top Slab (A, 2" thick):** 42" to **44"** (**2"** thick)

2. Master Cut List (All Parts Labeled A–R)

IMPORTANT: **Dado Allowance Included:** All shelf lengths (`C`, `D`, `E`, `F`, `H1`) include an extra **`1/2"`** (`1/4"` on each end) to seat into the `1/4"` deep dados in the Left & Right Side Panels (`B`). Similarly, the Vertical Dividers (`G`) include an extra **`1/2"`** (`1/4"` top and bottom) to seat into Shelves `C` and `D`.

Carcass & Top (2", 3/4", and 5/8" Quarter-Sawn Fir)

Part ID	Part Name	Qty	Thickness (T)	Width (W)	Cut Length (L)	Decimal (T × W × L)	Where It Goes & Joinery Notes
A	**Top Slab**	1	2"	19"	44 1/2"	2.000" × 19.000" × 44.500"	Sits on top of carcass (42"). Overhangs front & sides by 1", flush at back. Attach with figure-8 fasteners or elongated screw holes through H1.
B	**Outer Side Panels** *(Left & Right)*	2	5/8"	18"	42"	0.625" × 18.000" × 42.000"	Left and right outer walls. Route 1/4" deep cross-dados on inside faces for C, D, E, and stopped/vertical grooves for F & H1.
C	**Bottom Shelf** *(Flush w/ Floorboard)*	1	5/8"	16 3/4"	41 3/4"	0.625" × 16.750" × 41.750"	Sits between Front & Rear Floorboards (F) from 1 7/8" to 2 1/2" height so its top face is dead flush with the top of F. Route two 1/4" dados on top face for G.
D	**Lower Full Shelf** *(Above 3 Bays)*	1	5/8"	18"	41 3/4"	0.625" × 18.000" × 41.750"	Separates the 12 5/8" 3-bay section from the 10 1/2" middle section. Route two 1/4" dados on its **underside** for dividers G.
E	**Upper Full Shelf** *(Heavy-Duty 3/4")*	1	**3/4"**	18"	41 3/4"	0.750" × 18.000" × 41.750"	Separates the 10 1/2" middle bay from the 15" top bay. Milled to 3/4" to prevent sag across the 41 1/4" open span.
F	**Front & Rear Floorboards**	2	5/8"	2 1/2"	41 3/4"	0.625" × 2.500" × 41.750"	One at bottom front, one matching at bottom back (locks the open-back base against racking). Top edge is flush with top of Shelf C at 2 1/2".
G	**Lower Vertical Bay Dividers**	2	5/8"	18"	13 1/8"	0.625" × 18.000" × 13.125"	Divides lower 12 5/8" bay into three sections: **13 1/4" Left**, **13 1/2" Center (Drawers)**, and **13 1/4" Right**.
H1	**Top Rear Anti-Racking Cleat**	1	5/8"	2"	41 3/4"	0.625" × 2.000" × 41.750"	Mounted flush at the top back of the 15" upper bay (right under Top A) to prevent side-to-side wobble and anchor Top A.
H2	**Rear Drawer Stop Stretcher**	1	5/8"	2"	13 1/2"	0.625" × 2.000" × 13.500"	Mounted at the back of the 13 1/2" center bay on Shelf C so both drawers stop flush with the front face.

Center Bay Drawers & Shop-Made Runners (Builds 2 Drawers)

SHOP TIP: **How the Shop-Made Runner System Works (`1/4" × 1/2"` Runner `R`):** Each runner (`R`) is **`1/4"` thick × `1/2"` tall × `17"` long** and is glued/screwed to the inside face of Vertical Dividers `G`. Each Drawer Side (`K`, `5/8"` thick) has a **`1/4"` deep × `17/32"` wide groove** plowed down the center of its outside face. When installed, `3/16"` of the runner rides inside the drawer groove and `1/16"` spaces the drawer box off the divider wall—giving smooth action and allowing your inset drawer front (`J1`) to be the exact same width (`13 3/8"`) as the drawer box!

Part ID	Part Name	Qty	Thickness (T)	Width / Height (W)	Cut Length (L)	Decimal (T × W × L)	Where It Goes & Joinery Notes
J1	**Inset Drawer Fronts**	2	5/8"	6 1/8"	13 3/8"	0.625" × 6.125" × 13.375"	Front faces of Upper & Lower drawers. Gives a uniform 1/8" reveal (penny gap) at top, middle, and bottom (6 1/8" + 6 1/8" + 3×(1/8") = 12 5/8"). Cut 3/8" wide × 3/8" deep rabbets on inside ends to receive sides K.
J2	**Drawer Backs**	2	5/8"	5 3/8"	12 5/8"	0.625" × 5.375" × 12.625"	Rear walls of Upper & Lower drawer boxes. Captures 1/4" into dados near the rear of Drawer Sides K, sitting above Bottom L.
K	**Grooved Drawer Sides**	4	5/8"	5 7/8"	17"	0.625" × 5.875" × 17.000"	Left & right sides for both drawers. Plow a **1/4" deep × 17/32" wide groove** along the outside face (centered 2 11/16" up from bottom edge) for Runner R.
L	**Drawer Bottom Panels**	2	3/8"	12 5/8"	16 1/4"	0.375" × 12.625" × 16.250"	Resawn 3/8" fir (or plywood) captured in 1/4" deep × 3/8" wide grooves cut 3/8" up from the bottom inside edges of J1 and K.
R	**Shop-Made Drawer Runners**	4	1/4"	1/2"	17"	0.250" × 0.500" × 17.000"	Straight-grain quarter-sawn fir or hardwood strips. Mount inside Dividers G: lower pair bottom edge at **5 5/16"** from floor (2 13/16" above Shelf C); upper pair bottom edge at **11 9/16"** from floor (9 1/16" above Shelf C).

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3. Router Dado Layout Guide (Where to Cut Each Groove)

Use this checklist when laying out your 1/4" deep dados so every piece lines up on assembly:

A. Inside Faces of Left & Right Side Panels (B — Measure up from Bottom Edge = 0")

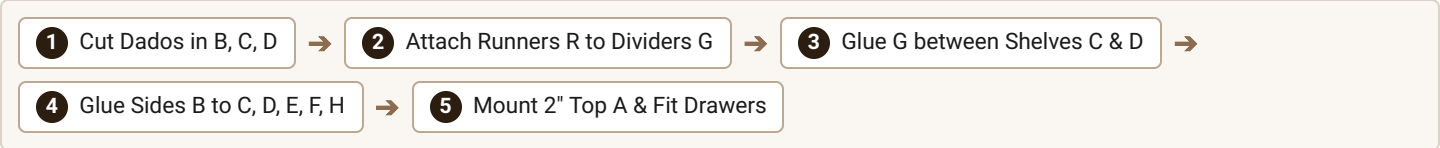
- **Front & Rear Floorboard Dado (F):** 0" to 2 1/2" height along the front 5/8" edge and back 5/8" edge (1/4" deep × 5/8" wide).
- **Bottom Shelf Dado (C): 1 7/8" to 2 1/2"** from the bottom edge (1/4" deep × 5/8" wide, running between front and rear floorboards).
- **Lower Full Shelf Dado (D): 15 1/8" to 15 3/4"** from the bottom edge (1/4" deep × 5/8" wide, full 18" width).
- **Upper Full Shelf Dado (E): 26 1/4" to 27"** from the bottom edge (1/4" deep × **3/4" wide**, full 18" width).

B. Top of Bottom Shelf (C) & Underside of Lower Shelf (D — Measure in from Left End of Board)

(Remember: 1/4" of each end of C and D seats inside Side Panels B) * **Left Vertical Divider Dado (G): 13 1/2" to 14 1/8"** from the left end of the shelf (1/4" deep × 5/8" wide). *(Leaves 13 1/4" clear left bay).* * **Right Vertical Divider Dado (G): 27 5/8" to 28 1/4"** from the left end of the shelf (1/4" deep × 5/8" wide). *(Leaves 13 1/2" clear center drawer bay and 13 1/4" clear right bay).*

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4. Recommended Shop Assembly Order



1. **Dry-Fit Sub-Assembly First:** Cut the dados in B, C, and D. Before gluing the outer case, glue and screw the four **Drawer Runners (R)** onto the inside faces of the two **Vertical Dividers (G)**—it is much easier to measure and clamp them flat on your workbench than inside the assembled cabinet.
2. **Assemble the Lower Core (C + G + D):** Glue the two **Vertical Dividers (G)** between the **Bottom Shelf (C)** and **Lower Full Shelf (D)**, checking that the center bay is square (13 1/2" W × 12 5/8" H).
3. **Bring the Sides Together (B + E + F + H):** Lay one **Side Panel (B)** flat on the bench, glue in the C/G/D assembly, the 3/4" **Upper Shelf (E)**, the **Front & Rear Floorboards (F)**, and the **Rear Cleats (H1, H2)**, then cap with the second **Side Panel (B)** and clamp square.
4. **Attach Top (A) & Fit Drawers (J/K/L):** Fasten the 2" **Top Slab (A)** flush at the back with a 1" overhang on the front and sides. Build the two drawer boxes (J1, J2, K, L), wax the wooden runners (R) with paste wax or paraffin, and slide the drawers home!